

Program and Branch Name:
 [BTech. CSE]

PBL Progress Report – I

Field	Details
Course Name	Data Structure and Algorithms - II
Course Code	CCSE0301
Faculty Name	Mr. Shamshad Ali
Assignment Type	Individual Assignment
Student(s) Name	Amaan Ansari
Roll No.	2501330100052
Team/Group No. (Group/Capstone only)	Not Applicable — Individual Assignment
Problem Statement / Project Title	Digital Certification Verification & Management System
SDG(s)	SDG 4 – Quality Education
Reporting Month	Month 1
Date of Submission	16/09/2026

Student Declaration

I certify that this assignment is my own work, with all sources appropriately acknowledged. I understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarized or inappropriately AI-generated.

Student Name & Signature: Amaan Ansari



Date: 10/09/2026

PBL Progress Report – I

Review 1: Problem Understanding & Initial Progress

Review Coverage: Individual Assignment: Unit/Module 1 (Trees) and Unit/Module 2 (Graphs)

Field	Details
1. Problem Statement / Assignment or Project Title	Digital Certification Verification & Management System This Digital Certification Verification & Management System is a conceptual system designed to manage and verify digital certificates issued by authorized institutions or organizations. The system will allow authorized issuers to maintain certificate records and enable users or verifiers to search and verify certificate information using a unique certificate ID or other relevant details.
2. Problem Context & Background	Digital certificates are commonly used to represent achievements, qualifications, training, courses and other forms of certification. When certificate records are maintained manually or distributed across different systems, verifying a certificate can become time-consuming. A verifier may need to contact the issuing organization or manually search through many records. As the number of certificates increases, the system needs an efficient method to organize, store and search certificate information. A structured data management approach can help improve the retrieval of certificate records. The proposed system explores the application of Data Structures and Algorithms-II concepts for organizing certificate records and representing relationships between certificates, issuers and recipients.
3. Problem Identification / Understanding	Based on the study done during Month 1, the following understanding has been developed: ● Manual verification of certificates can be time-consuming. ● Finding a particular certificate can become difficult when the number of certificate records increases. ● Duplicate or incorrectly entered certificate information can make management difficult. ● The system requires an efficient method for searching certificate records using a certificate ID or other key. ● Certificate records may contain information such as certificate ID, recipient name, issuing organization, course/program, issue date and status. ● Different certificates may be associated with different issuing organizations and categories. ● The system should provide an organized structure for storing and retrieving certificate information. At this stage, no part of the Digital Certification Verification & Management System has been built, this section only reflects problem understanding.
4. Objectives	● To develop a conceptual Digital Certification Verification & Management System. ● To make searching and retrieving certificate records easier and more organized.. ● To explore the use of Binary Search Tree (BST) and AVL Tree for efficient certificate searching. ● To explore the use of Heap for priority-based management of certificate-related tasks such as pending verification or renewal. ● To explore Graph representations using Adjacency Matrix and Adjacency List for representing relationships between certificate issuers, recipients and certificates.
5. Target Users / Stakeholders	● Students / Certificate Holders – manage and present their certificates. ● Educational Institutions – issue and maintain certificate records. ● Training Organizations – issue course/training certificates. ● Employers / Recruiters – verify certificates submitted by applicants. ● Verification Authorities – verify certificate information.
6. Research / Literature Review Completed	A research paper, “Automated Certificate Verification and Management System” by Bahalkar P., Shingade P., Gavhane N., Khatri V., Pote A., & Chilakapati S. (2024), was studied to understand existing approaches for digital certificate verification and management. The paper discusses automated verification using QR codes, embedded verification links and certificate information extraction. Limitations: LMS integration, blockchain implementation, real-time alerts, and multilingual support were identified as future improvements. https://healthinformaticsjournal.com/index.php/IJMI/article/view/1103 Another research paper, “Academic Certification Validation System: A Web-Based Framework for Secure Digital Verification of Academic Credentials” by Akanksha Vyas Gupta and Asst. Prof. Dr. Meena Mashru (2026), was studied to understand existing approaches for digital academic certificate verification and management. The paper discusses a web-based system using centralized certificate storage, unique certificate codes, role-based access control, certificate issuance, and real-time verification. Limitations: Dependence on a centralized database, limited entropy in certificate-code generation, and the need for blockchain-based tamper evidence and mobile support were identified as areas for future work. https://zenodo.org/records/19950613 These papers helped in understanding the practical requirements of academic certificate management and provided background for identifying suitable DSA-II concepts for the proposed system.

7. Unit/Module Concepts Identified for Application	DSA Concept	Possible Application / Purpose	Time complexity
	Binary Tree	Organizing certificate categories and hierarchical information	O(n) traversal
	BST	Efficient searching of certificate records using Certificate ID	O(log n) avg, O(n) worst case for search/insert
	AVL Tree	Maintaining balanced certificate records for efficient searching	O(log n) for search/insert/delete
	Tree Traversal	Systematic processing/display of certificate records	O(n)
	Max/Min Heap	Prioritizing certificate requests according to defined priority	O(log n) insert/delete, O(1) find min/max
	Graph	Representing relationships between issuers, recipients and certificates	Depends on representation and O(V + E) for BFS/DFS traversal
	Adjacency Matrix	Representing relationships between a relatively fixed set of entities	O(V ²) space, O(1) edge lookup
	Adjacency List	Efficiently representing relationships when connections are comparatively sparse	O(V+E) traversal, O(k) edge lookup
<i>Note: The concepts above have been identified as proposed/possible applications at this stage; none of them have been implemented yet. Implementation is planned for later reviews.</i>			
8. Work Completed During Month 1	- Identified and finalized the project problem statement. - Identified major stakeholders such as certificate holders, issuing organizations and certificate verifiers. - Identified possible applications of BST and AVL Tree for certificate searching. - Identified possible use of Heap for priority-based certificate management. - Identified possible use of Graphs for representing relationships between issuers, recipients and certificates. - Identified Adjacency Matrix and Adjacency List as possible graph representations for the proposed system. - Planned the next stage of system design and implementation.		
9. Evidence / Supporting Material	The main study material used for this work was the DSA-II notes and PPTs shared by the faculty. These helped in understanding the concepts covered in Trees, BST, AVL, Heap, Graphs, Adjacency Matrix and Adjacency List. NIST SP 800-63-4 was referred to for understanding the digital identity, authentication and credential-management context. It is background/security guidance, not the DSA implementation itself. Research Papers <i>Automated Certificate Verification and Management System</i> (2024) <i>Academic Certification Validation System: A Web-Based Framework for Secure Digital Verification of Academic Credentials</i> (2026) Github link - https://github.com/amaan-28/Digital-Certificate-Verification-System		
10. Challenges Faced	- Understanding how Tree data structures can be practically applied to certificate records. - Deciding which DSA concepts are genuinely useful for the proposed system instead of unnecessarily using every concept from the syllabus. - Deciding whether BST or AVL Tree would be more suitable for certificate searching. - Understanding how certificate IDs and other attributes should be organized for efficient searching. - Understanding how issuers, certificate holders and certificates can be represented as graph entities. - Understanding the practical difference between Adjacency Matrix and Adjacency List.		
11. Individual / Team Contribution	As this is an Individual Assignment, all the work reported below was carried out by me alone: - I identified and defined the problem statement for the Digital Certification Verification & Management System. - I studied DSA-II Units 1 and 2 (Trees and Graphs) from the course material. - I mapped the relevant DSA concepts to the digital certificate management and verification problem at a conceptual level. - I studied the two research paper for better understanding of this project.		
12. Overall Progress (%)	25% This percentage reflects the completion of the initial problem understanding, requirement analysis, DSA-II concept study (Units 1 and 2), and preliminary conceptual planning for the Digital Certification Verification & Management System. The actual design, application of the identified concepts, and implementation are planned for the subsequent review stages.		
13. Plan for Next Review	- Define the certificate record structure and important fields. - Decide the key used for certificate searching, such as Certificate ID. - Design the initial BST/AVL Tree structure for certificate records. - Design a basic priority structure using Heap for pending verification/management tasks. - Design the relationship model between certificate issuers, holders and certificates. - Compare Adjacency Matrix and Adjacency List for representing these relationships. - Implement the initial data structures. - Test basic insertion, searching and retrieval operations. - Report the outcomes in Review 2.		